

A smaller-threshold tail consequence for Bernoulli sections

Run `fa_banach_001`, arXiv:math/0601369, literature-implied partial subcase

Source Question

Artstein-Avidan, Friedland, Milman and Sodin study a $p \times n$ Bernoulli sign matrix X , with $p = (1 - \delta)n$, and

$$S = n^{-1}XX^T.$$

Their Theorem 2 proves

$$\mathbb{P}\{\lambda_{\min}(S) \leq \delta^2/8\} \leq \exp(-cn^{1/6}\delta)$$

under the stated lower bound on δ . Immediately after comparing this with the Litvak–Pajor–Rudelson–Tomczak-Jaegermann estimate, they ask whether this left-hand side is in fact as small as

$$\exp(-n\delta^C/C)$$

for some universal $C > 0$ [1].

Supporting Literature

Dabagia and Fernandez prove a rectangular smallest-singular-value lower-tail theorem for random matrices with independent entries and bounded $2 + \beta$ moments. In particular, for an $N \times m$ matrix A in their setting, $N > m$, their theorem gives constants $C, c > 0$ such that

$$\mathbb{P}\left\{\sigma_m(A) \leq \varepsilon(\sqrt{N+1} - \sqrt{m})\right\} \leq (C\varepsilon)^{N-m+1} + e^{-cN}$$

for all $\varepsilon > 0$ [2]. Bernoulli sign entries satisfy the moment hypotheses with universal constants.

Implied Partial Answer

Theorem 1. *There are universal constants $c_0, c_1, c_2 > 0$ such that, for every $0 < \delta < 1/2$, $p = (1 - \delta)n$, and $p \times n$ Bernoulli sign matrix X ,*

$$\mathbb{P}\left\{\lambda_{\min}(n^{-1}XX^T) \leq c_0\delta^2\right\} \leq e^{-c_1\delta n} + e^{-c_2n}.$$

Proof. Apply the Dabagia–Fernandez theorem to $A = X^T$, viewed as an $n \times p$ matrix. Choose a fixed $\varepsilon_0 > 0$ so small that $(C\varepsilon_0)^{n-p+1} \leq e^{-c_1(n-p+1)}$. Then

$$\mathbb{P}\{\sigma_p(X) \leq \varepsilon_0(\sqrt{n+1} - \sqrt{p})\} \leq e^{-c_1(n-p+1)} + e^{-c_2n}.$$

Since $p = (1 - \delta)n$ and $0 < \delta < 1/2$,

$$\sqrt{n+1} - \sqrt{p} \geq \sqrt{n} - \sqrt{(1 - \delta)n} = \sqrt{n}(1 - \sqrt{1 - \delta}) \geq \frac{\delta\sqrt{n}}{2}.$$

Thus the event

$$\lambda_{\min}(n^{-1}XX^T) = \sigma_p(X)^2/n \leq \varepsilon_0^2\delta^2/4$$

is contained in the event estimated above. Set $c_0 = \varepsilon_0^2/4$. □

Status and Limitations

This is a literature-implied partial subcase. It gives the exponential tail shape requested in the 2006 question only after replacing the source threshold $\delta^2/8$ by a smaller universal multiple $c_0\delta^2$. The exact left-hand side in the source question, with threshold $\delta^2/8$, is not claimed to be answered here.

The supporting paper does not explicitly cite arXiv:math/0601369 as the question being answered; the implication is obtained by identifying the source's covariance eigenvalue with the square of the smallest singular value of the transposed rectangular Bernoulli matrix.

References

- [1] S. Artstein-Avidan, O. Friedland, V. Milman, and S. Sodin, *Polynomial bounds for large Bernoulli sections of ℓ_1^N* , arXiv:math/0601369, 2006.
- [2] M. Dabagia and M. Fernandez, *The smallest singular value of inhomogenous random rectangular matrices*, arXiv:2408.14389, version dated 2026.